

RESIDUAL STRENGTH AFTER IMPACT AT CURVED EDGES

Part: **1.4 – COMPOSITE MATERIALS**
Chapter: **1423**
Title: **RESIDUAL STRENGTH AFTER IMPACT AT CURVED EDGES**
Revision: **0.2**
Date: **23/01/2024**

1 INTRODUCTION

1.1 OBJECT

The aim of this chapter is to define and present a method to obtain the residual strength in laminated plates after an impact at or near a curved edge. The low velocity edge impact is that caused by a blunt object with a trajectory normal to the plate at the curved edge or in the plane of the plate (where ever such case is possible). The BVID threshold of visibility for this case is conveyed, within Airbus, to be defined for the Detailed Visual Inspection (DVI) as that resulting in an edge crack of 30mm, or a dent depth of 1mm after transverse impact, or a dent depth of 0.6mm after edge impact (0.3mm after relaxation, see Ref. 5 or Ref. 6).

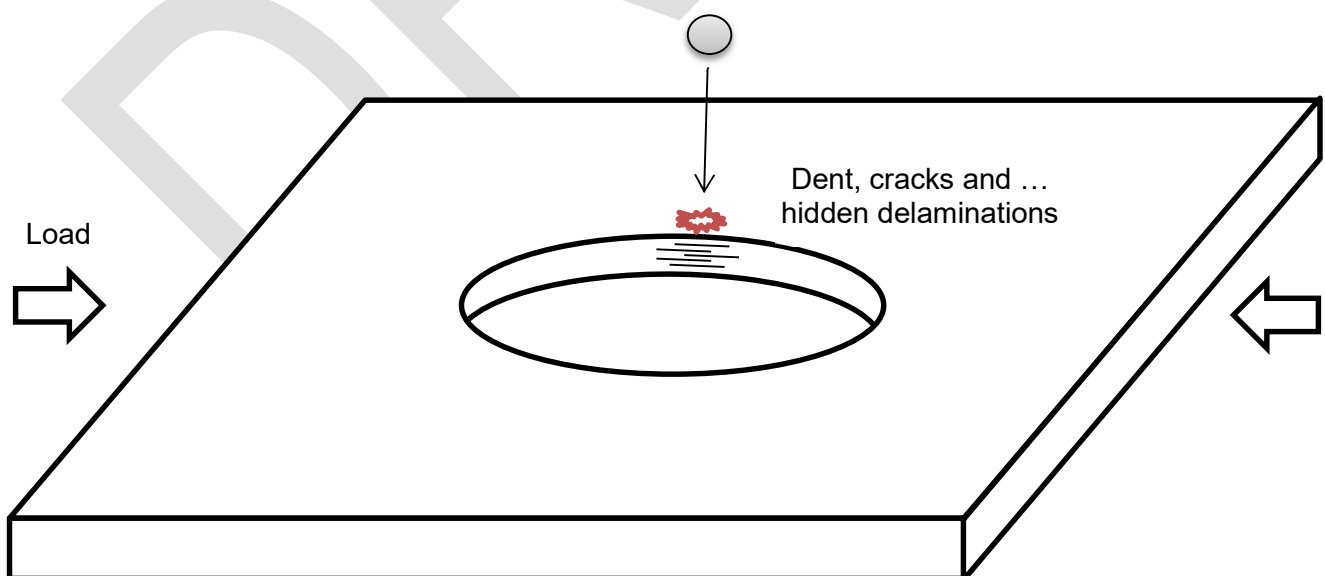


Figure 1. Sketch of the CAIH problem (residual strength after impact in curve edges)

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1.2 SCOPE AND LIMITATIONS

Method applicable to:

- ✓ Fibrous composite solid laminates (made of UD-tape or biaxial-fabrics)
- ✓ Stacking sequences following the typical design rules
- ✓ Low velocity impacts causing BVID damages
- ✓ Curved edges, or hole edges, of plates

Limitations:

- ✓ The method proposed relies on an correction of the residual strength at the straight edge (same laminate) with the curvature radius of the edge. Thus, the strength at the straight edge must be known (see chapter 1422).

2 METHOD DESCRIPTION

2.1 Generalities

See discussion in chapters 1421 “RESIDUAL STRENGTH AFTER TRANSVERSE IMPACT IN COMPOSITES” and 1422 “RESIDUAL STRENGTH AFTER EDGE IMPACT IN COMPOSITES”.

2.2 Tension After Impact at/near Hole edge or curved edge (TAIH)

Strength after impact in tension at curved edges (*TAIH*) is not expected to be a sizing driver in general. The maximum of the design values available for compression (*CAIH*) or tension strengths after transverse impact (*TAI*) or edge impact (*TAIE*) can be conservatively used if no dedicated allowable is available.

2.3 Compression After Impact at/near Hole edge or curved edge (CAIH)

The compression strength of a fibrous composite laminate after an impact at a curved edge or near a curved edge (transverse impact), at the BVID threshold (DVI), so called *CAIH* within this document, is defined as the theoretical peak tangential strain reached at the curved edge in the undamaged/pristine material at failure loads level. The following approximation model is suggested, using the residual strength after impact on straight edges (*CAIE*) as reference, and a correction factor function of the curvature radius of the edge:

$$CAIH = K_C \cdot CAIE$$

This methodology is inspired in Airbus Commercial method (Ref. 1 and Ref. 2), but the curvature correction is derived from the analytical solution of the stress distribution around holes (in absence of sufficient experimental data to feed an empirical regression formula). The characteristic distance

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approach proposed by Whitney and Nuismer (Ref. 3) will be used to obtain the correction curve for curvature.

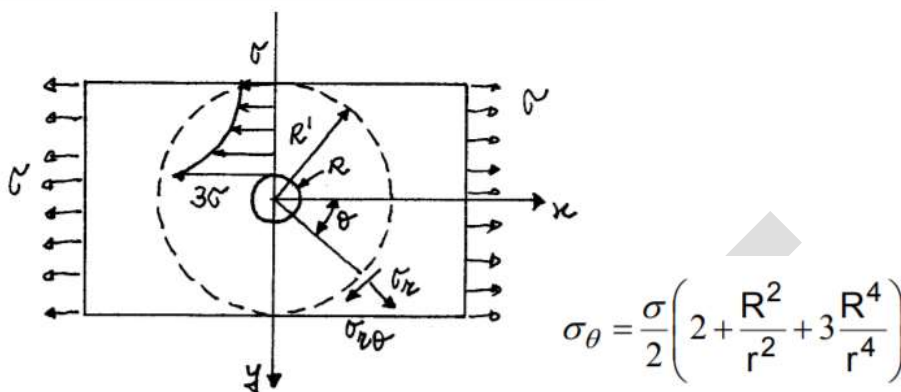


Figure 1. Stress at distance from hole edge

The assumption made is the existence of a generic value of a characteristic distance $a0$, of the same order of the BVID typical size, which can be used to refer the residual strength (as strain):

$$\varepsilon_{(a0)} = Kt_{(r=R+a0)} \cdot \varepsilon_{gross}$$

$$\varepsilon_{peak} = Kt_{(r=R)} \cdot \varepsilon_{gross} \quad \rightarrow \quad \varepsilon_{peak} = \varepsilon_{(a0)} \cdot Kt_{(r=R)} / Kt_{(r=R+a0)}$$

Defining $CAIH$ as the theoretical peak strain on edge at failure load, as for straight edges the Kt will be 1.0 either at the edge or at the characteristic distance $a0$, the ultimate $\varepsilon_{(a0)}$ must be $CAIE$ (provided the assumptions made are right (*)), then:

$$CAIH = CAIE \cdot Kt_{(r=R)} / Kt_{(r=R+a0)} = CAIE \cdot 3 / \{ 0.5 \cdot [2 + (R/r)^2 + 3(R/r)^4] \} \quad \dots \text{with } r = R + a0$$

(*) Dedicated testing to be done. Validation should rarely cover low edge radii (of the order of BVID). The approximation model shall be truncated for radius below R equal to 50mm. Fitting this model (at $R = 50mm$) with that proposed by Airbus Commercial for one particular CFRP in Ref. 2 it shown to be some more conservative than that of Airbus Commercial (see following plot).

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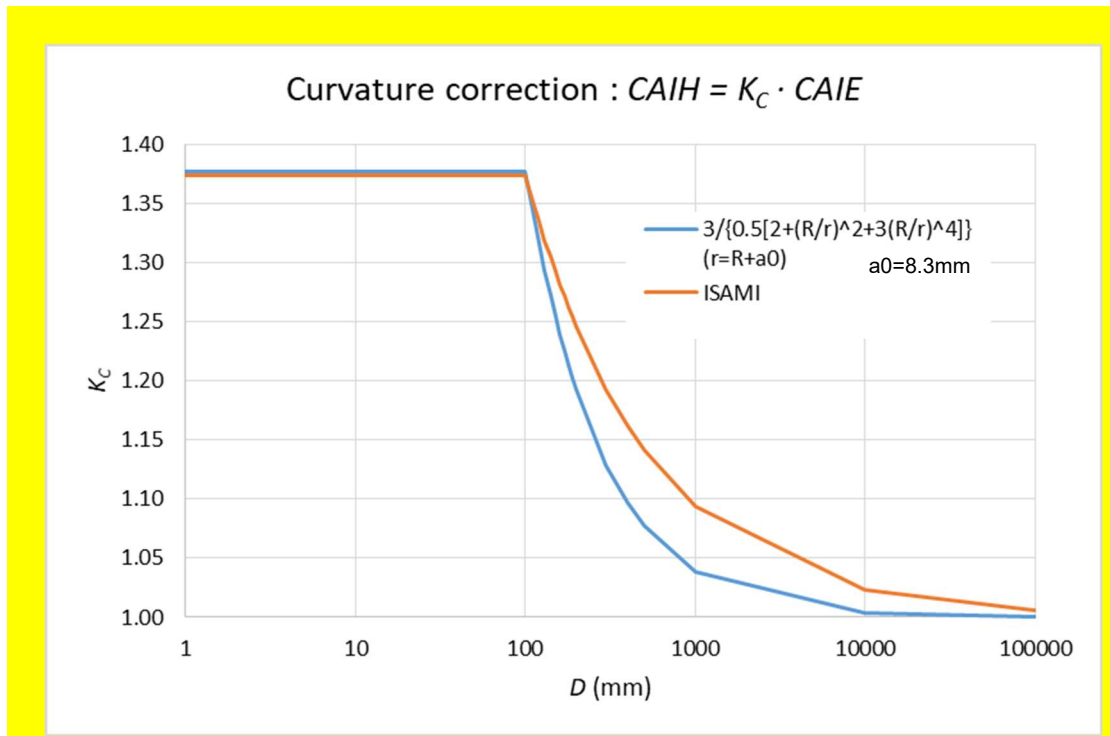


Figure 2. Correction factors for CAIE with curvature (tentative $a_0=8.3\text{mm}$)

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3 APPENDICES

TESTS

As discussed in previous pages, the design value for residual strength after impact on a curved edge (and the function proposed to capture the influence of the curvature radius) must be validated by dedicated tests.

Very few tests data are available within Airbus DS for this topic (see Ref. 4), and indeed no standard test method is known to be proposed here. The following figure sketches possible tests configurations to consider for this purpose. Test specimens and fixture shall be designed to prevent buckling occurs before the strength failure. FEM analysis, properly calibrated, shall be typically required to correlate failure loads with the theoretical peak strain values at damage location. Design values for the BVID threshold at edges (see chapter §1422) should cover both edge impact and transverse impact near the curve edge.

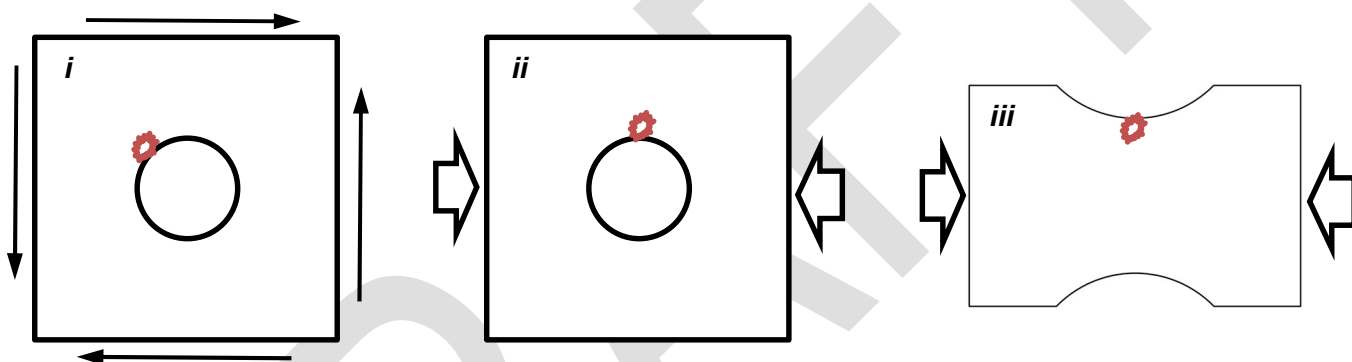


Figure 3. Possible test arrangements for CAIH (shear i and compression ii, iii)

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
BVID	-	Barely Visible Impact Damage, either determined by the visibility threshold, or by the impact energy cut-off. The detectability threshold for impacts near edges is conveyed (within Airbus) to be set for the Detail Visual Inspection (DVI), as 1.0mm dent depth after impact (0.3mm after relaxation), or 30mm long edge crack. On the other hand, the energy cut-off is typically set in 35 J, in general function of the part location in the airframe.
CAIH	$\mu\epsilon$	Residual Compression strength After Impact, at curved edge or near a curved edge (transverse impact), in a fibrous composite laminate including a BVID. Normalized as the theoretical peak tangential strain reached at the curved edge in the undamaged/pristine material, at failure loads.
CFRP	-	Carbon Fibre Reinforced Polymer/Plastic
DET/DVI	-	Detailed Visual Inspection
E	MPa , J	Elastic (Young's) Modulus or impact energy, from context

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<i>EKDF</i>	-	Environmental knock down factor, i.e. the correction factor to apply to a property obtained at room temperature in order to get the value at applicable hot/wet conditions
<i>FEM</i>	-	Finite Element Model
<i>K_B</i>	-	B-basis knock down factor (ratio B-basis to mean)
<i>TAI</i>	MPa , $\mu\epsilon$	Residual Tension strength After transverse Impact in a fibrous composite laminate including a BVID
<i>TAIE</i>	MPa , $\mu\epsilon$	Residual Tension strength After Edge Impact in a fibrous composite laminate including a BVID (straight edge)

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[Airbus] V57RP1627379	Is. 1.0	A350XWB-2000 Stressing and Design Criteria for LAGRANGE MDO
Ref. 2	[Airbus] V57RP1039013	Is. 4.0	Edge Impact Damage Tolerance Method for A350
Ref. 3	[J.M.Whitney, R.J.Nuismer]	1974	Stress Fracture Criteria for Laminated Composites Containing Stress Concentrations
Ref. 4	TEA-CS-NT-190030	Is. A	Test on spar web panels. Analysis of results.
Ref. 5	[Airbus] ME0905804	Is. 1.0	PRELIMINARY UPDATE TO BVID REQUIREMENTS FOR EDGE IMPACTS ON A350 XWB COMPOSITE STRUCTURES
Ref. 6	[Airbus] AITM1-0076	Is. 1	Airbus Test Method Fibre. Reinforced Plastics. Determination of compression and tension strength after edge impact

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

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RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.2 23/01/2024	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Structural Analysis</i>	-	Florian Glock ??/01/2022
<i>Stress Methods & Optimization</i>	G. Baños 23/01/2024	-

Table 3. *Approval signatures table for last revision of this chapter*